

\$ 2 million

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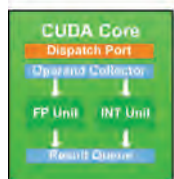
NVIDIA Ion 2

Acer has announced that their Aspire One 532G will feature an Ion 2 graphics solution

How stuff works

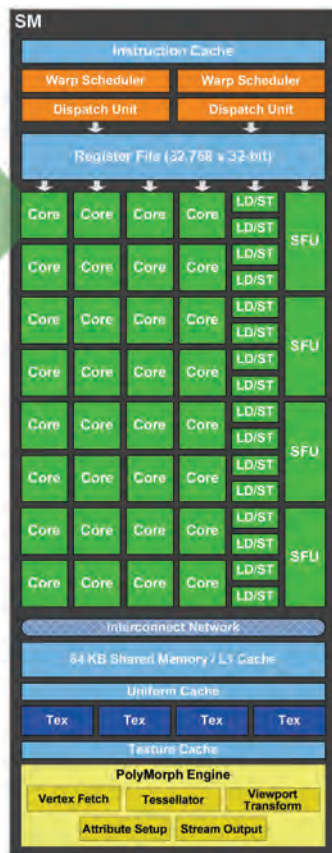
or fused-multiply-add. Rounding off of decimal places is done in a single step and there is no loss in precision in this single stage step, but there is a saving of processing resources – win-win. There are four SFUs or Special Function Units within an SM that handle transcendental operations such as sine, cosine, square root and reciprocal.

The improved cache on the GF100 is another point NVIDIA has illustrated on, and the most apparent benefit is from the much larger 768 KB unified L2 cache. The benefits of a unified cache are improved utilisation, less resource utilisation and therefore higher efficiency. Incidentally, all processing on the GF100 is IEEE-754 2008-compliant – this simplifies developers' lives as well as keeps them honest. Additionally, double precision floating point performance has gone up by 400 per cent compared to the GT200.



The emphasis on geometry: but why?

DX11 is a lot more rigid than DX10 when it comes to rendering techniques and features; therefore both GPU makers have more of a level playing field than ever since neither can add extra features on their GPU without the fear of it being unused, thereby wasting super precious die space. DX11 makes Tessellation a mandatory feature and this has been fully exploited by NVIDIA, whereas rivals AMD,



The Streaming Multiprocessor

they claim, have only scratched the surface with their RV870. Thanks to its geometry prowess the scaling down with Tessella-

tion will be minimal on the GF100

and this means complex and lifelike characters, objects, environments and scenery with more fine detail than the competition using the same game assets. Environments should look a whole lot better. Environment interaction will be more of an experience and deformation of objects will become more realistic. Tessellation needs oodles of spare geometry power, since triangle density of a given frame can scale up to astronomic proportions, putting a sudden, enormous load on the geometry sub system and it is this feature that has dictated the reworking of the Raster Engine and the creation of the PolyMorph Engine.

The emphasis on geometry: how?

The PolyMorph Engine is an improved version of the Geometry Controller on the GT200. Unlike the GT200 and predecessors, each SM gets its own PolyMorph Engine.

SPECIFICATIONS COMPARISON

	Fermi aka GF100 (GTX 480)	GT200 (GTX 285)	RV 870 (HD 5870)
Stream Processors / Cores	512	240	320 (1600 ALU / 5)
Core Clock (MHz)	unknown	648	850
Stream processor clock (MHz)	unknown	1476	850
Memory Clock (MHz)	unknown	2484	4400
Texture address (mapping) units	64	80	80
Texture filtering per clock *	256 (4 x 64)	80 (1 x 80)	80 (1 x 80)
ROPs (Raster operator units)	48	32	32
Fabrication process	40nm	55nm	40nm
Memory bus width	384-bit	512-bit	256-bit
Memory type	GDDR5	GDDR3	GDDR5
Memory size	unknown	1 GB	1 GB

* in the GF100 each texture mapping unit can fetch four texture samples per clock, as opposed to one sample on other cards

	GT200	GF100	Benefits
L1 Texture Cache (per quad)	12KB	12 KB	Fast texture filtering
Dedicated L1 LD/ST Cache	None	16 or 48 KB	Efficient physics and ray tracing
Total Shared Memory	16 KB	16 or 48 KB	More data reuse among threads
L2 Cache	256 KB (Texture read only)	768 KB (all clients read/write)	Greater texture coverage, robust compute performance

Previously, each TPC was made up of three SMs and had a single Geometry Controller. So from sitting inside a pipeline, the PolyMorph Engine becomes modular and all 16 of them can intercommunicate, thereby work cohesively, effectively doing more, more efficiently. The PolyMorph Engine has five stages as shown below. Results after each stage are passed to the connected SM, where it is executed, and the result is passed to the next stage in the PolyMorph Engine.

The Raster Engine comprises hardware that is basically responsible for creating screen pixels from the geometry data received from the SPs and the Polymorph Engine and this data is passed back to the SPs for further pixel shading. The GF100 has four Raster Engines, one each for each GPC. The Raster Engine is unchanged from the GT200 as far as we know.

These changes give the GF100 a massive boost to geometry crunching – 8x the power of the GT200. Geometry performance has stagnated, from neglect more than anything; the GT200 was only 3x faster than the NV30 (FX 5800) in this regard.